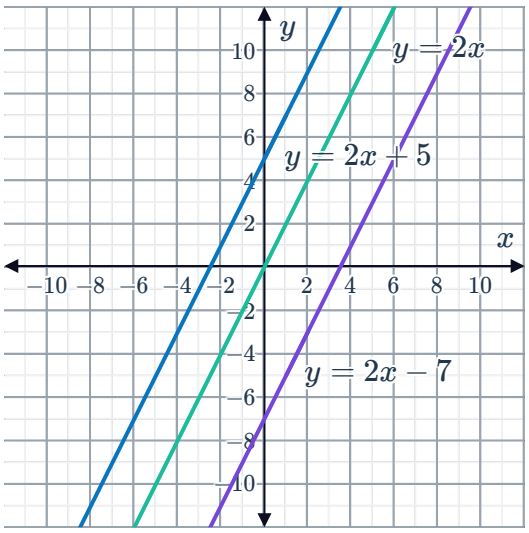
Parallel lines

1 The equations $y = 2x$, $y = 2x + 5$ and $y = 2x - 7$ have been graphed on the same number plane:



a What do all of the equations have in common?

b What do all the lines have in common?

2 State whether the following pairs of lines are parallel:

a $y = -3x - 2$ and $y = -3x + 9$

b $y = -2x - 5$ and $y = -2x - 8$

c $y = 7x + 8$ and $y = -5x + 8$

d $y = 4x - 1$ and $y = 4x - 6$

e $x = 4$ and $y = 5$

f $y = 7x - 5$ and $y = -7x + 6$

g $y = 3x - 5$ and $y = -\frac{1}{3}x + 6$

h $y = -8x - 2$ and $9x + 7$

3 State whether the following lines are parallel to $y = 7x + 3$:

a $y = 7x - 3$

b $y = 6x + 3$

c $y = 7x + 4$

d $y = 7x$

e $y = 6x + 4$

f $y = -7x + 3$

g $y = 3x + 7$

h $y = -3x - 7$

4 State whether the following lines are parallel to $y = -3x + 2$:

a $y = 3x$

b $-3y - x = 5$

c $y = -10 - 3x$

d $y + 3x = 7$

5 State whether the following lines are parallel to $y = 9x + 2$:

a $y = 9x$

b $y = -9x + 5$

c $y = -9x + 2$

d $y = 9x - 2$

6 Consider the lines $L_1 : 5x + 8y - 10 = 0$ and $L_2 : y = -\frac{5x}{8} + 8$.

a Find the gradient of line L_1.

b Find the gradient of line L_2.

c Are the two lines parallel?

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7 Consider the line $y = 2x + 2$. If every point on the line is shifted 2 units up, find the equation of the new line.

8 Every point on a particular line is shifted 6 units down. The resulting line has equation $y = 2x - 2$. Find the equation of the original line.

9 Find the equation of a line described by the following information:

a Parallel to the x-axis and passes through $(-10, 2)$.

b Parallel to the y-axis and passes through $(-7, 2)$.

c Parallel to the line $y = -3x - 8$ and cuts the y-axis at -4.

d Parallel to the line $y = 8x - 3$ and cuts the y-axis at 5.

e Parallel to the line $y = -2x + 9$ and passes through the point $(-3, 1)$.

10 If the following pairs of lines are parallel, find the value of b :

a $0 = 5x - 9y + 6$
$0 = bx + 3y - 2$

b $y = bx + 4$
$2y = \frac{4}{5}x - 7$

11 The line L_1 goes through $(3, 2)$ and $(-2, 4)$.

a Find the gradient of line L_1.

b Find the equation of the line that has a y-intercept of 1 and is parallel to line L_1.

12 The line L_1 goes through $(-9, 10)$ and $(2, -1)$.

a Find the gradient of line L_1.

b Find the equation of the line that passes through $(10, 8)$ and is parallel to line L_1.

13 The line L_1 passes through the point $(9, -5)$ and is parallel to the line $y = -5x + 2$.

a Find the gradient of line L_1.

b Find the equation of line L_1.

Perpendicular lines

14 Describe what it means for two lines to be perpendicular.

15 If two lines are perpendicular, state the product of their gradients.

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16 State whether the following pairs of lines are perpendicular:

a $y = x + 1$ and $y = x - 1$

b $y = 3x - 7$ and $y = -3x + 6$

c $y = -3x + 6$ and $y = \frac{x}{3} + 7$

d $y = \frac{2x}{3} + 4$ and $y = -1 - \frac{3x}{2}$

e $y = -\frac{x}{3} - 9$ and $y = -(3x - 27)$

f $y = 3x - 5$ and $y = -\frac{1}{3}x + 6$

g $y = -\frac{2x}{3} + 1$ and $y = -1 + \frac{3x}{2}$

h $y = \frac{3x}{5} + 6$ and $y = \frac{-5x - 30}{-3}$

17 Given that the following pairs of lines are perpendicular, calculate the value of m:

a $5x + 4y + 8 = 0$
$mx + 9y - 8 = 0$

b $-6x + 6y - 8 = 0$
$mx + 10y + 8 = 0$

18 Find the gradient of the line perpendicular to the following lines:

a Line with gradient 6

b $y = 6x + 3$

c $y = \frac{x}{8} + 3$

d $y = 8 - x$

19 Consider the lines L_1: $8x = -7y + 3$ and L_2: $7y - 8x - 7 = 0$.

a Find the gradient of line L_1.

b Find the gradient of line L_2.

c Are the two lines perpendicular?

20 Find the equation of a line described by the following information:

a Perpendicular to the x-axis and passes through $(-8, -1)$.

b Perpendicular to the y-axis and passes through $(-8, -8)$.

c Perpendicular to $y = -\frac{x}{2} + 5$, and goes through the point $(0, 6)$.

d Perpendicular to $y = 6x + 10$, and has the same y-intercept.

e Perpendicular to the line that passes through $(1, 1)$ and $(3, 13)$, and has a y-intercept of $(0, -4)$.

21 The line L_1 is perpendicular to $y = 5x - 4$ and cuts the y-axis at 1.

a Find the gradient of line L_1.

b Find the equation of line L_1.

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22 The line L_1 is perpendicular to $y = 10x + 8$ and passes through the point of intersection of the lines $y = x + 5$ and $8x - 10y + 60 = 0$.

a Find the gradient of line L_1.

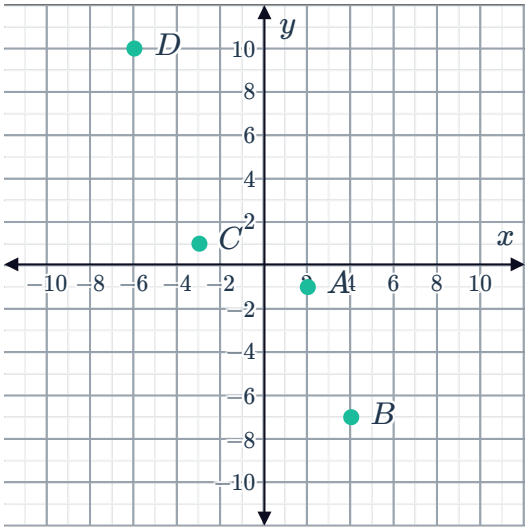
b Find the point of intersection of $y = x + 5$ and $8x - 10y + 60 = 0$.

c Find the equation of the perpendicular line L_1.

Applications

23 Consider the following points on the number plane:

- Point $A\,(2, -1)$
- Point $B\,(4, -7)$
- Point $C\,(-3, 1)$
- Point $D\,(-6, 10)$



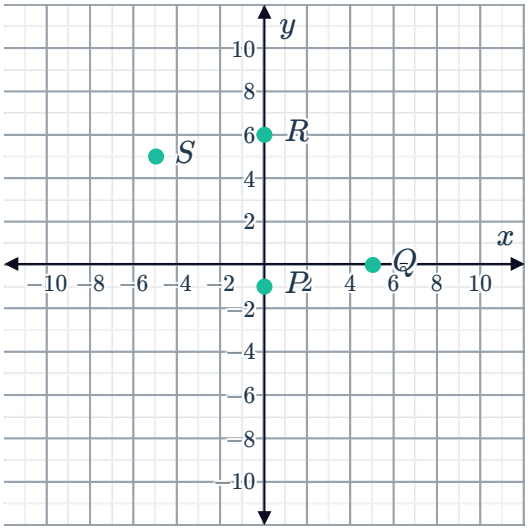
a Find the gradient of the line AB.

b Find the gradient of the line CD.

c Is the line CD parallel to AB?

24 Consider the following points on the number plane:

- Point $P\,(0, -1)$
- Point $Q\,(5, 0)$
- Point $R\,(0, 6)$
- Point $S\,(-5, 5)$



a Find the gradient of PQ.

b Find the gradient of RS.

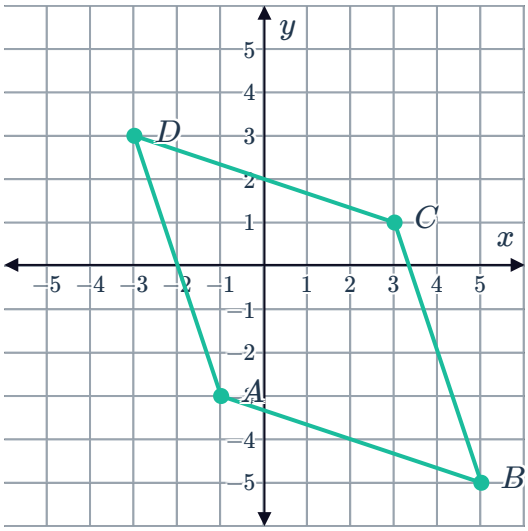
c Are PQ and RS parallel?

d Are QR and PS parallel?

e What type of quadrilateral is $PQRS$?

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25 Consider the rhombus $ABCD$ on the number plane:



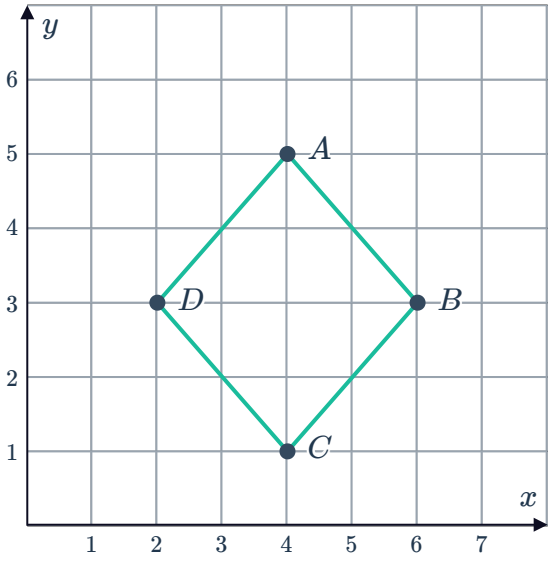
a Find the gradients of its diagonals.

i Gradient of AC

ii Gradient of BD

b Are the diagonals of the rhombus perpendicular? Explain your answer.

26 The four vertices of quadrilateral $ABCD$ have been plotted on the number plane:



a Find the gradient of the following sides:

i AB **ii** BC **iii** DC **iv** AD

b Given that side $AB = BC$, determine the type of quadrilateral described by the four points.

27 Determine whether the following sets of points are collinear:

a Point $A\,(-4, 3)$, Point $B\,(-2, 7)$ and Point $C\,(-7, -3)$.

b Point $A\,(-3, 1)$, Point $B\,(-2, 6)$ and Point $C\,(0, 14)$.

c Point $A\,(-2, 4)$, Point $B\,(2, 2)$ and Point $C\,(4, 1)$.

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